

Rept. Bk. No. 26-103

DEPARTMENT OF MINES
SOUTH AUSTRALIA

MAGNETIC SURVEYS IN THE MIDDLEBACK RANGES

October - December 1948

Geophysical Report 2/49 D.M. 62/49

by

C. Kerr Grant

DEPARTMENT OF MINES
SOUTH AUSTRALIA

MAGNETIC SURVEYS IN THE MIDDLEBACK RANGES.

I. INTRODUCTION.

For about six weeks towards the end of 1948, geophysical observations were undertaken together with geological work in the Middleback Ranges, South Australia, for the purpose of ascertaining to what extent geophysical work could be used for the location of further deposits of ore, iron and to assist in the elucidation of the general geological structure of the area. It had been known for some time that the area was magnetically disturbed, and consequently magnetic observations only were carried out, in the hope of being able to elucidate the magnetic anomalies. The area does not, on cursory examination seem particularly suitable for other types of geophysical work. The observations were undertaken in part also to investigate the suitability of the area for an airborne magnetometer survey. Some work of this nature has since been carried out.

II AREA INVESTIGATED -

Firstly the Iron Prince Hill was selected as the most suitable area to find out the possible correlation of magnetic observations with known bodies of iron ore; the geology of this area has been well established and the Hill is free from the magnetite-bearing haematite quartzite formations which form such a prominent feature of the rest of the Middleback Ranges. The observations at the Iron Prince were carried out as a series of traverses along the lease boundaries of the area with subsequent observations along offsets at right angles to these traverses.

Secondly, two traverses were also observed.

along the southern boundary of the Iron Queen and South Iron Baron leases in order to ascertain the variations of magnetic intensity over large masses of hematite - quartzite.

Thirdly in order to ascertain the regional magnetic variation outside the ranges, a long magnetic traverse, with stations a quarter to half a mile apart was carried out along the Kimba Road.

III TECHNIQUE

Observations were initially carried out with Watts vertical magnetic force variometer No. 46898. This instrument was made available to the Department by Enterprise Exploration Co. Pty. Ltd, of Broken Hill. The instrument had been adjusted to a sensitivity of approximately 14 gammas per scale division by W.G. Fenner of this Department for his observation in the Wallaroo-Moonta area. This high sensitivity was unnecessary for the observations and some time could have been saved by using a lower sensitivity as the instrument is somewhat sluggish at high sensitivity but having commenced observations with this sensitivity, it was decided not to alter it; the average distance apart of stations in the area was 200 feet over rough ground, and very little actual economy in time would have been achieved. On 27th November 1948 the Department took delivery of a new vertical magnetic force variometer No. 60565, and observations were commenced with this instrument which had a sensitivity of nearly 30 gammas per scale division.

For the purpose of obtaining accurate Drift curves most of the stations in the area were repeated. The magnitude of the magnetic anomalies obtained subsequently proved that this refinement was unnecessary.

The sensitivity of both instruments was checked every two or three days by standardization with their auxiliary magnets, and by interchanging these magnets in different instruments, the magnets were checked against each other. During the

period of the observations, that is from 22nd October, until 9th December, the sensitivity of instrument No. 46898 changed in sensitivity from 14.1 to 14.3 gammas per scale division; and instrument No. 60565 from 29.0 to 29.3 from 29th November until 9th December; over the same periods the constants of the auxiliary magnets changed with respect to one another by less than half of one per cent, and the constants of the magnets of instrument 60565, calibrations of which had been supplied by Greenwich Observatory, agreed in ratio to the same accuracy.

The average drift in a morning or afternoon between repeated stations was approximately 10 gammas, but after correction for drift the error on normal readings should be less than 10 gammas except for very large readings. As the anomalies even on the Iron Prince Hill are of the order of several hundred gammas, it is evident that no appreciable instrumental errors have been introduced.

The errors arising from superficial magnetic effects on the ground are more difficult to estimate, and are quite large in the most anomalous areas, up to fifty or a hundred gammas, but tests made in areas of slight variation indicate that they are in general less than ten to fifteen gammas. The effect on the results from superficial magnetic effects is thus also negligible.

IV RESULTS -

The vertical Magnetic Intensity over the Iron Prince Hill is shown in Fig 1, which also shows the known outcrops of haematite ore. (A geological plan of the same area, obtained through the courtesy of the Broken Hill Proprietary Co. Ltd. is shown in Fig. 1. A blue areas represent haematite ore). It is seen that there is concentration of the magnetic contours over the ore, with magnetic highs on the north and east side, and magnetic lows on the south, with a maximum variation of just over 1000 gammas. The included areas of amphibolite produce further complexity of the

contours as they break the magnetisation of the main ore body thus forming an extra magnetic high and low at the boundaries of the amphibolite. The ore body is delineated to approximately the accuracy to be expected from stations 200 ft. apart, that is to about + feet. It should be possible to delineate the limits of the ore considerably more accurately by more closely spaced stations, possibly to within + 20 feet. Haematite itself is almost non-magnetic and limonite only slightly magnetic, so the magnetic effects from the haematitic ore bodies are presumed to arise from the presence of small particles of magnetite included in the ore.

In order to observe how the vertical magnetic intensity was affected by depth inside the body, Magnetic profiles are observed across the top and inside tunnel 1A; these are shown in Fig(2) together with the geological section of the hill along the line. Little extra information can be obtained from these results, and since they can only be observed in localities where underground workings exist their application is very limited. It can perhaps be inferred that the maximum depth of ore should be between 400 and 700 feet from the eastern end of the tunnel.

The two east-west profiles along the south end of the Iron Queen and South Iron Baron leases are shown in Fig (3), together with annotations on the surface geology along the southern traverse. Phenomally large variations in magnetic intensity were obtained over the uptilted haematite quartzite strata, the highest peaks being over 50,000 gammas in magnitude. It is evident that it would not be easy to pick out in such areas small anomalies of the order of 1,000 gammas obtained from ore bodies. The geology shows fair general correlation with the magnetic profile; it is considered that much better correlation could be obtained from a regional magnetic survey expressed as a contoured plan.

Fig. (4) shows the variation of magnetic intensity along the Kimba Road which runs from Whyalla to Kimba through

the Kimba Gap in the Middleback Ranges. The traverse extends from twelve miles east of the Kimba Gap to four miles west. One of the geological problems of the Middleback Ranges is whether or not the haematite profile does not give sufficient information to decide this point. The relatively small magnetic anomalies produced in the gap may be accounted for either by the effect of large magnetic poles produced at the South End of the iron Warrior on the north side of the Gap, and at the north side of Cook's lease and the North Iron Knight, on the South side of the Gap, or by the effect of deep seated haematite quartzite, 100 feet or more below the surface in the gap. It is certain that the haematite quartzite does not lie immediately below the surface in the Gap. It is considered that by making further traverses through the Gap, and across the ends of the range near the Gap, the absence or otherwise of haematite quartzite under the gap could be demonstrated. A cursory examination of the geology along this traverse was made by Dr. K.R. Miles and reasonable correlation occurs between geological features and magnetic anomalies. As the stations are a quarter to half a mile apart, many minor features in the magnetic traverse may have been missed. The most interesting feature of the traverse is the anomaly at its extreme west end; this is in line with the geometrical extension of the Camel Range, which contains similar haematite quartzite formations to the main Middleback Range. The magnetic profile indicates that these formations must persist under the surface as far south as the Kimba Road. The surface in this area consists of limestone ridges, sand dunes and wind blown deposits of seed gypsum.

V CONCLUSIONS

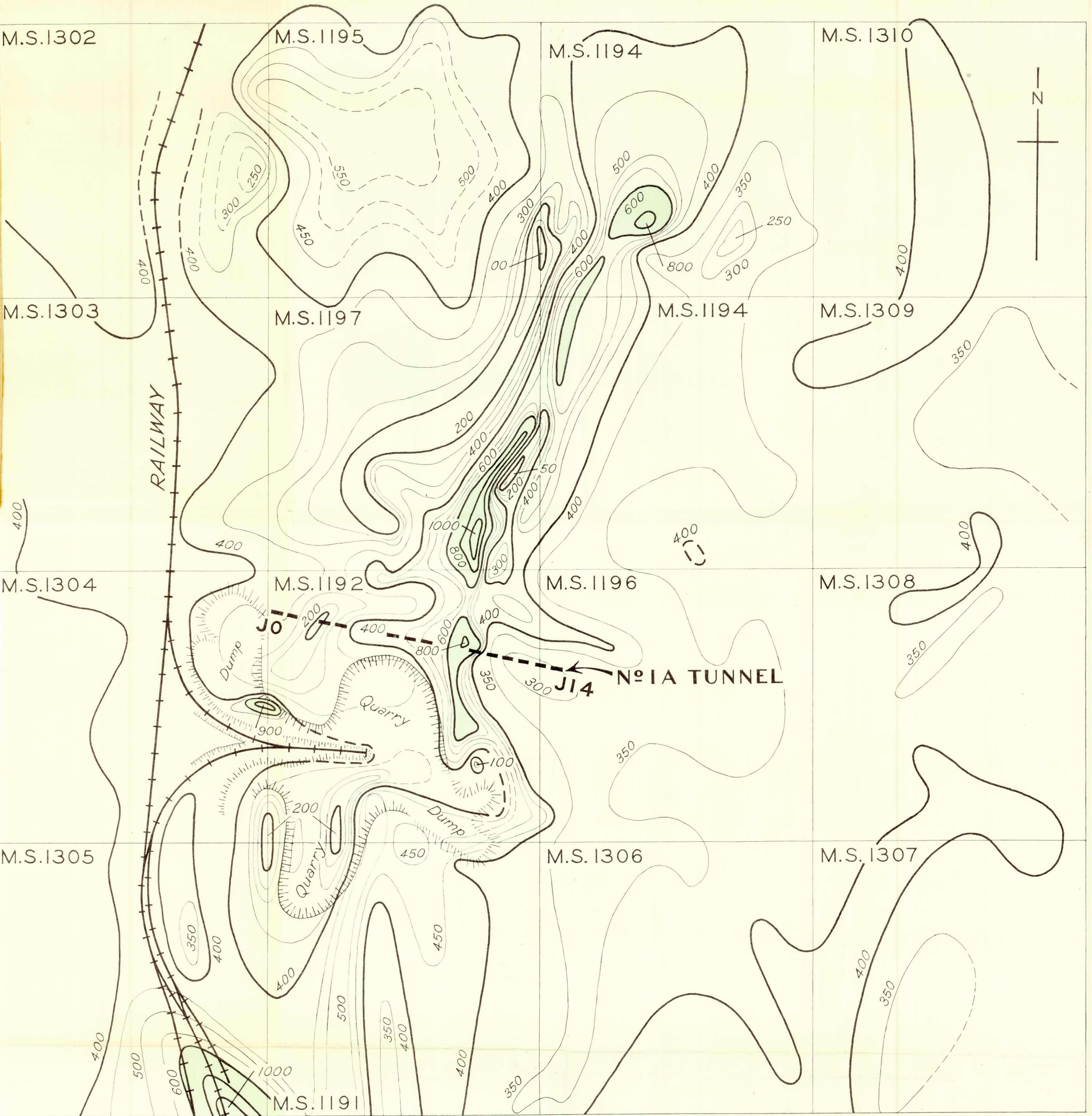
It is considered that a magnetic survey over proved haematitic ore bodies can be used to delineate the approximate limits of the ore and to indicate the presence of bodies of amphibolite or other non-magnetic or slightly magnetic minerals

included in it.

With present knowledge it is not considered possible to use magnetic observation to locate ore bodies in areas containing large masses of haematite-quartzite formations.

Magnetic methods, including also at times aerial magnetic methods, can be used with advantage to interpret geological structure in the Middleback Ranges by tracing haematite quartzite; still more so to find and trace similar buried structures outside the area of the ranges, such as the extension of the Camel Range.

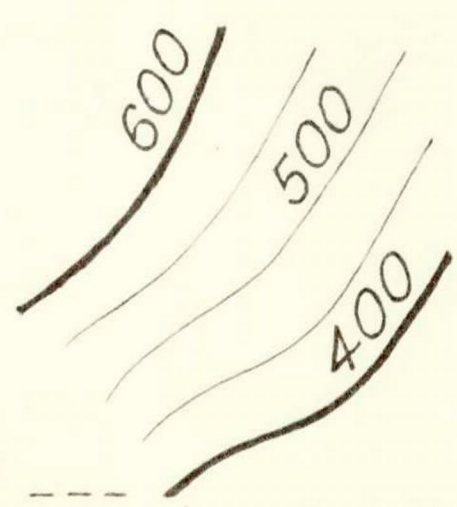
PLAN 31	L 50-33	FIG. 1.
	50-290	" 2
	L 50-46	" 3
	L 50-34	" 4



CHAINS 5 0 5 10 15 20 25 30 35 40 CHAINS

MAGNETIC CONTOURS

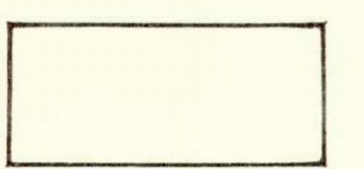
Contours at 50 gamma intervals from 00-600
 " " 200 " " " 600-1,000



MINERAL SECTION..... M.S. 1302

MAGNETIC HIGHS

Areas over 600 gammas



el. B.C.B. L50-33

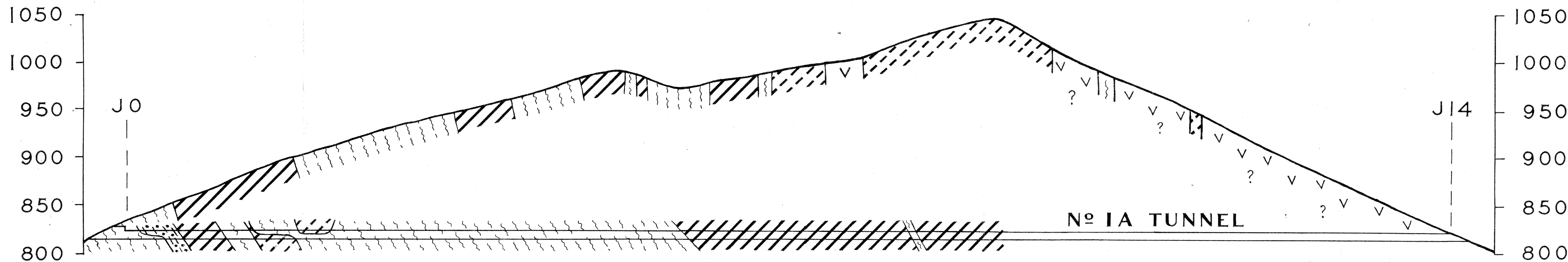
C. Kerr Grant **KG** for Geophysicist

Revised this line to 27 1/2 ins.

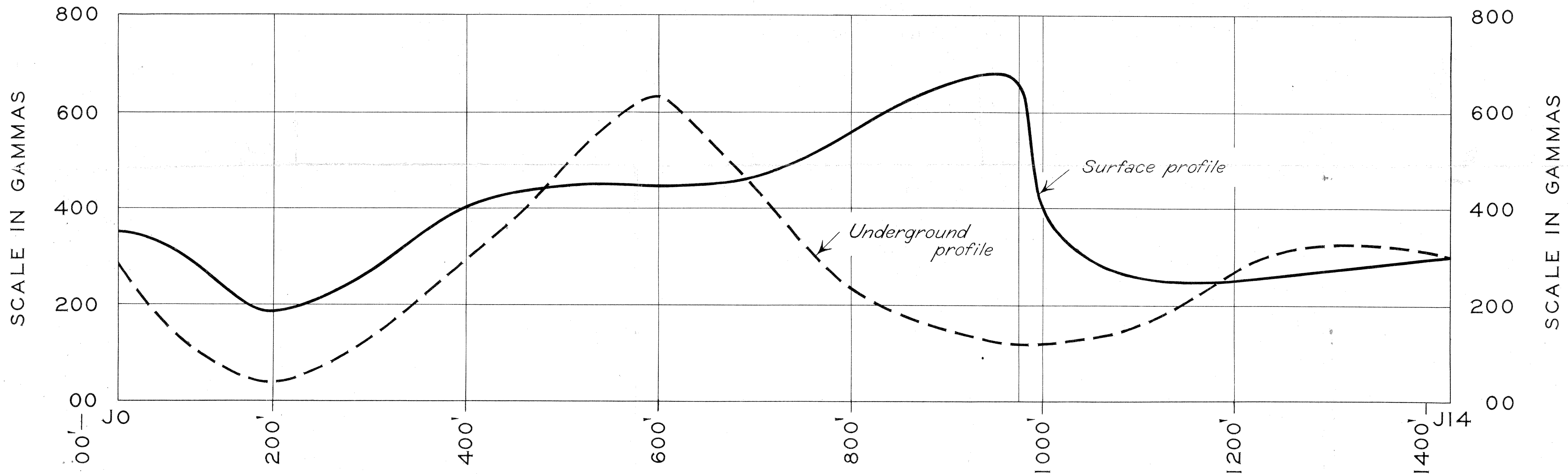
FIG. I

S. A. G. DEPT. OF MINES											
Key No. 1711		MIDDLEBACK RANGES, IRON PRINCE HILL VERTICAL MAGNETIC INTENSITY					Approved		Scale 200 feet to 1 inch		
Compiled from C. K. Grant original 49-65							1/4 Director of Mines	Drawn Tcd. <u>B.C.B.</u> Ckd. <u>[initials]</u> Evd.	Passed [initials] Chief Draughtsman	L50-33	Date 27-7-50 DE
C. Kerr Grant Senior Geophysicist											
[]											
[]											
[]											
[]											
[]											
[]											
[]											
[]											
[]											
[]											
[]											
[]											
[]											
[]											
[]											
[]											
[]											
[]											
[]											
[]											
[]											
[]											
[]											
[]											
[]											
[]											
[]											
[]											
[]											
[]											
[]											
[]											
[]											
[]											
[]											
[]											
[]											
[]											
[]											
[]											
[]											
[]											
[]											
[]											
[]											
[]											
[]											
[]											
[]											
[]											
[]											
[]											
[]											
[]											
[]											
[]											
[]											
[]											
[]											
[]											
[]											
[]											
[]											
[]											
[]											
[]											
[]											
[]											
[]											
[]											
[]											
[]											
[]											
[]											
[]											
[]											
[]											
[]											
[]											
[]											
[]											
[]											
[]											
[]											
[]											
[]											
[]											
[]											
[]											
[]											
[]											
[]											
[]											
[]											
[]											
[]											
[]											
[]											
[]											
[]											
[]											
[]											
[]											
[]											
[]											
[]											
[]											
[]											
[]											
[]											
[]											
[]											
[]											
[]											
[]											
[]											
[]											
[]											
[]											
[]											
[]											
[]											
[]											
[]											
[]											

VERT. SCALE IN FEET



GEOLOGICAL SECTION



MAGNETIC PROFILES



Del. B.C.B. 50-290

C. Kerr Grant Senior Geophysicist

C. Kerr Grant
Senior Geophysicist

S. A. G. DEPT. OF MINES					IRON PRINCE MIDDLEBACK RANGES		Approved		Scales: As shown	
GEOLOGICAL SECTION & MAGNETIC PROFILES THROUGH NO 1A TUNNEL					Req. No. 1711 D.M. C. K. Grant original 50-103		Drn. B.C.B. Tcd. B.C.B. Ckd. B.C.B. Exd. B.C.B.		Passed Chief Draughtsm. Date 17-8-50 D.E.	
Associated Drawing	No.	No.	Authority	Date	Amendment					

